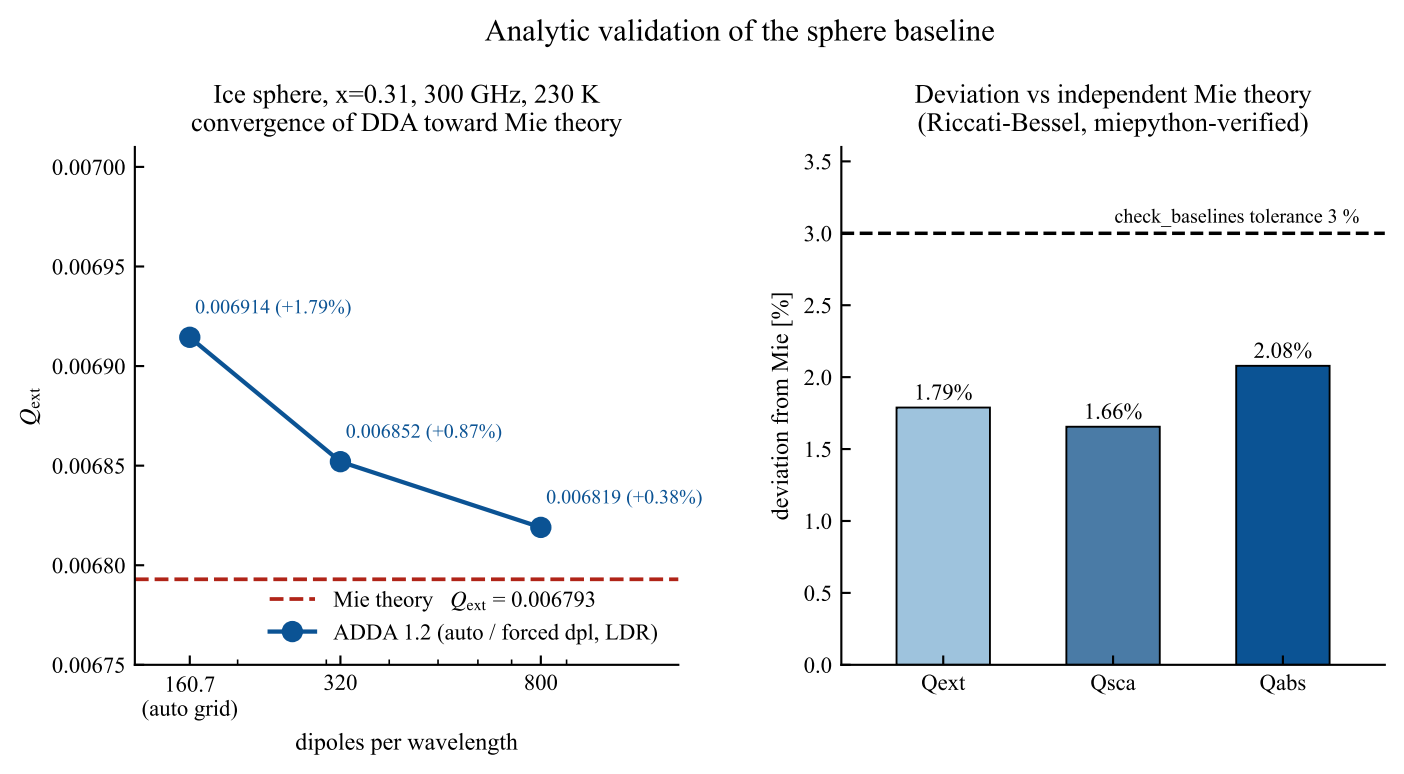
Analytic validation of the sphere baseline
Ice sphere, x=0.31, 300 GHz, 230 K
convergence of DDA toward Mie theory
0.00700
0.00695
0.00690
0.00685
0.00680
0.00675
Q_{ext}
0.006914 (+1.79%)
0.006852 (+0.87%)
0.006819 (+0.38%)
Mie theory Q_{ext} = 0.006793
ADDA 1.2 (auto / forced dpl, LDR)
160.7
(auto grid)
320
800
dipoles per wavelength
Deviation vs independent Mie theory
(Riccati-Bessel, miepython-verified)
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
deviation from Mie [%]
check_baselines tolerance 3 %
1.79%
1.66%
2.08%
Qext
Qsca
Qabs